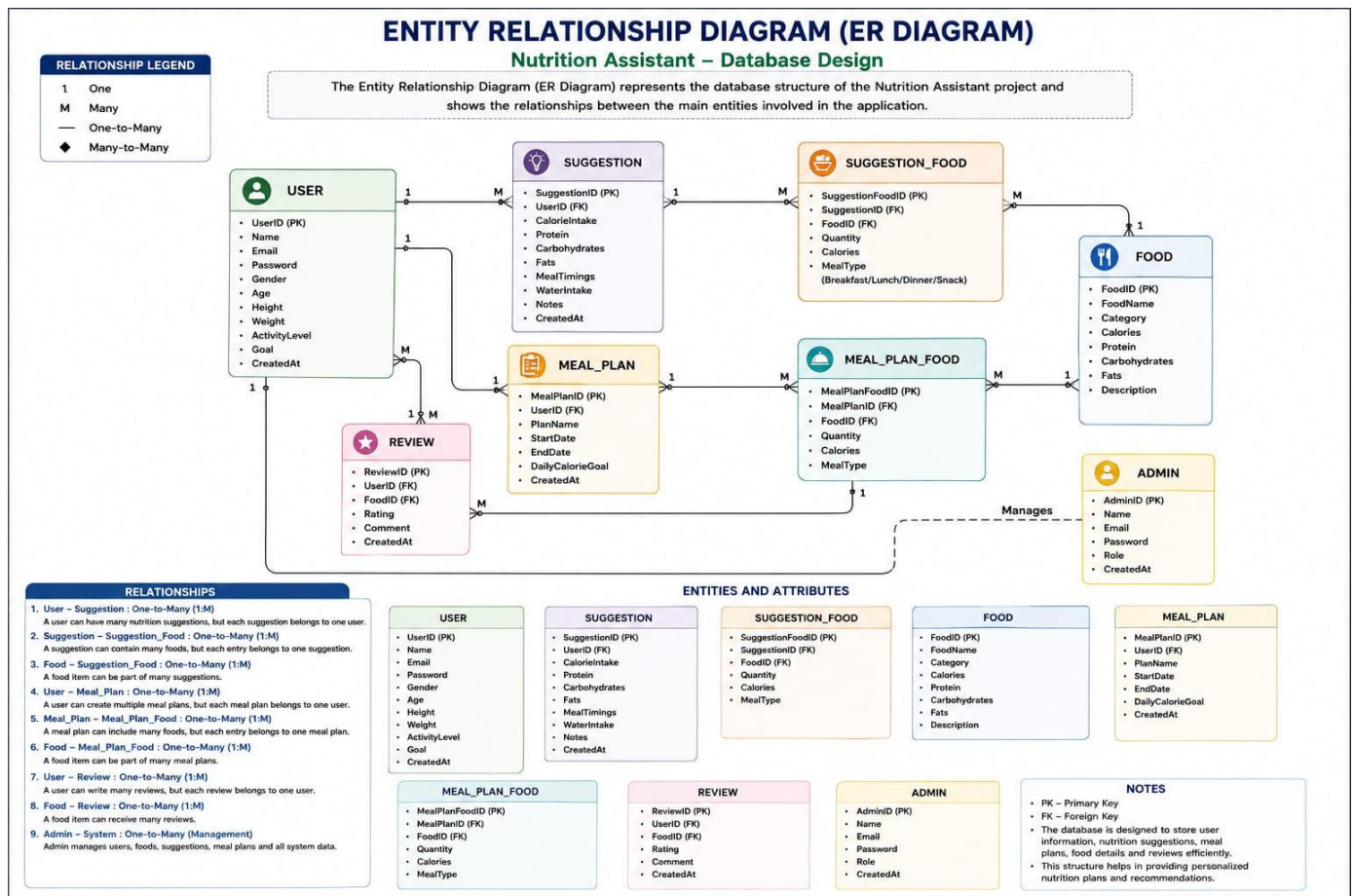


ENTITY RELATION DIAGRAM (ER DIAGRAM)

The **Entity Relationship Diagram (ER Diagram)** represents the database structure of the Nutrition Assistant application. It illustrates how the main entities interact with each other and how data is stored and managed in the MongoDB database. The application follows a One User → Many Nutrition Suggestions relationship, enabling users to create and manage multiple personalized nutrition plans.

Main Entities

- User
- Suggestion
- Admin



1. User – Suggestion Relationship

Type: One-to-Many (1:M)

A single user can generate multiple nutrition suggestions, but each nutrition suggestion belongs to only one user.

Implementation:

The Suggestion collection contains `userId` as a reference (`ObjectId`) to the User collection.

2. User – Authentication Relationship

Type: One-to-One (1:1)

Each registered user has one unique account secured using an email and encrypted password.

Implementation:

The User collection stores login credentials, while JWT tokens are used for authentication and authorization.

3. Admin – User Relationship

Type: One-to-Many (1:M)

The administrator manages multiple user accounts within the system.

Implementation:

The admin can view, monitor, and manage all registered users through the Admin Dashboard.

4. Admin – Suggestion Relationship

Type: One-to-Many (1:M)

The administrator can access, review, and delete nutrition suggestions generated by users.

Implementation:

Admin privileges allow complete management of nutrition suggestion records stored in MongoDB.

Main Entities and Attributes

- User
- UserID
- Name

- Email
- Password
- Age
- Gender
- Height
- Weight
- Activity Level
- Role
- Created At

Suggestion

- SuggestionID
- UserID (Reference)
- UserName
- Age
- Height
- Weight
- BMI
- Calorie Intake
- Protein Needs
- Carbohydrate Needs
- Suggested Foods
- Meal Timing
- Walking Recommendation
- Weight Goal
- Nutrition Suggestion
- Date

Admin

- AdminID
- Name
- Email
- Password
- Role